

Assignment 1

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1 Introduction

The aim of this assignment is to investigate the community structure of a network using different community detection algorithms.

The analyzed network represents similarities between anime extracted from the MyAnimeList dataset. Nodes represent anime, while edges represent similarity relationships based on shared genres. The dataset was filtered by considering only popular anime with at least 50,000 members and valid genre information.

Three community detection algorithms were applied: Louvain, Leiden, and Girvan–Newman. The obtained partitions were then compared using Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI) to evaluate the consistency between the detected communities.

2 Network Description

The graph was constructed from the filtered anime dataset. Each node represents an anime identified by its unique identifier. An undirected edge is created between two anime when they share at least two common genres.

The resulting network contains:

Property	Value
Number of nodes	3579
Number of edges	780393
Graph type	Undirected
Edge criterion	At least two shared genres

Table 1: Main properties of the anime similarity network.

Figure 1 shows the visualization of the anime similarity network.

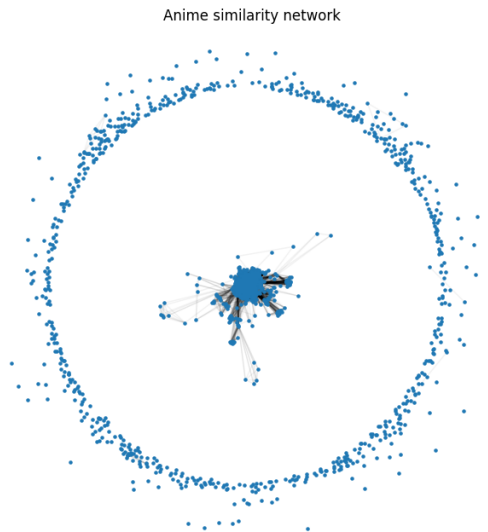


Figure 1: Visualization of the anime similarity network.

3 Community Detection

3.1 Louvain Algorithm

The Louvain algorithm was applied to the complete network. It detects communities by optimizing the modularity function, identifying groups of nodes with dense internal connections.

The algorithm detected 733 communities with an execution time of 5.099 seconds.

3.2 Leiden Algorithm

The Leiden algorithm was applied to the complete graph. It improves the Louvain approach by producing well-connected communities while maintaining modularity optimization.

The algorithm detected 734 communities with an execution time of 0.414 seconds.

3.3 Girvan–Newman Algorithm

The Girvan–Newman algorithm detects communities by progressively removing edges with high betweenness centrality.

Due to its computational complexity, the algorithm was applied only to a reduced graph composed of 200 randomly selected nodes.

The algorithm detected 48 communities with an execution time of 1.385 seconds.

Algorithm	Communities	Execution Time (s)
Louvain	733	5.099
Leiden	734	0.414
Girvan–Newman	48	1.385

Table 2: Comparison of community detection algorithms.

Louvain and Leiden produced a very similar number of communities on the complete graph, with 733 and 734 communities respectively. However, Leiden required significantly less execution time, indicating better computational efficiency in this experiment.

Girvan–Newman was applied to a reduced subgraph. Therefore, its number of detected communities cannot be directly compared with those obtained by Louvain and Leiden on the complete graph.

4 Community Size Analysis

The community sizes were analyzed for all three algorithms. The results show a highly uneven distribution, with a few large communities and many very small ones. Louvain and Leiden have very similar mean community sizes (4.882 and 4.876 nodes, respectively), while Girvan–Newman has a mean size of 4.166 nodes on the reduced 200-node graph. The ten largest communities for each algorithm are reported below.

Community	Size
0	1002
1	865
3	671
5	272
281	21
297	8
26	7
300	4
51	3
392	2

Table 3: Louvain

Community	Size
0	984
1	960
2	626
3	230
4	20
5	11
6	8
7	7
8	4
9	3

Table 4: Leiden

Community	Size
1	149
7	3
0	2
12	2
3	1
4	1
5	1
2	1
6	1
8	1

Table 5: Girvan–Newman

The community sizes range from 1 to 1002 nodes for Louvain, from 1 to 984 nodes for Leiden, and from 1 to 149 nodes for Girvan–Newman. In particular, the first four Louvain and Leiden communities are substantially larger than the remaining ones, highlighting the strongly skewed structure of the partitions.

5 Evaluation of Community Quality

Since no ground truth community assignment is available, the detected partitions were compared using Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI). These metrics measure the similarity between different community assignments without determining which partition is the correct one.

The comparison was performed between Louvain and Leiden, since both algorithms were applied to the complete graph.

Comparison	NMI	ARI
Louvain vs Leiden	0.9285	0.8625

Table 6: Similarity between Louvain and Leiden community partitions.

The obtained values ($\text{NMI} = 0.9285$ and $\text{ARI} = 0.8625$) indicate a strong agreement between the Louvain and Leiden partitions. Although the detected partitions are not identical, both algorithms identify a highly similar underlying community structure. Girvan–Newman was not included in this comparison because it was applied only to a reduced graph, whereas Louvain and Leiden were applied to the complete network.

6 Community Size Distribution

The distribution of community sizes was analyzed separately for the Louvain and Leiden partitions.

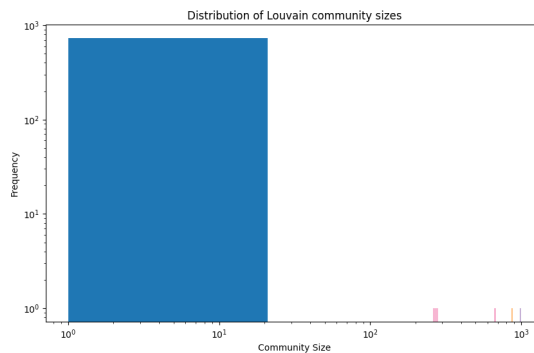


Figure 2: Distribution of Louvain community sizes.

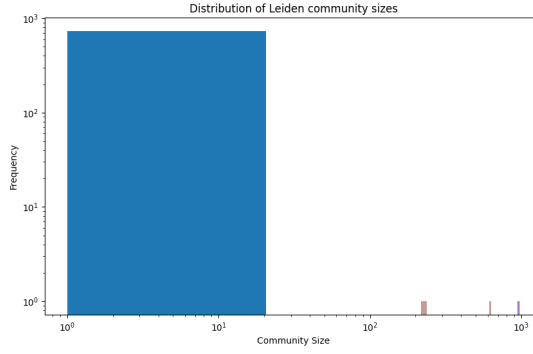


Figure 3: Distribution of Leiden community sizes.

Both distributions show a strongly uneven community structure, with a small number of very large communities and a large number of small communities. The four largest Louvain communities contain 1002, 865, 671, and 272 nodes, while the four largest Leiden communities contain 984, 960, 626, and 230 nodes.

7 Community Visualization

The detected communities were visualized by assigning different colors to the nodes according to their community label.

Figures 4, 5 and 6 show the partitions obtained with Louvain, Leiden and Girvan–Newman respectively.

For Louvain and Leiden, all detected communities are represented in the visualizations. Since hundreds of communities are present, the colormap provides different color values for the community labels, although individual colors cannot all be uniquely distinguished by eye.

7.1 Louvain Communities

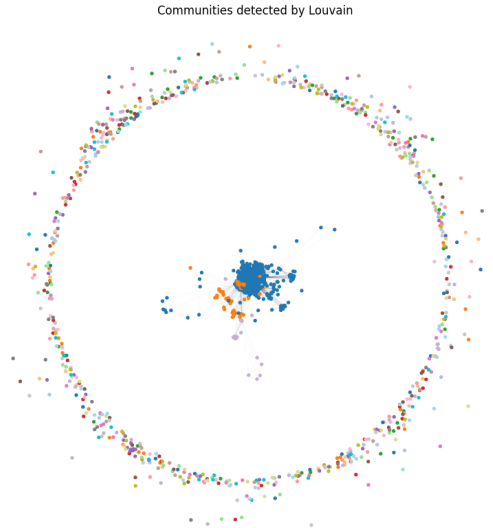


Figure 4: Communities detected by the Louvain algorithm.

7.2 Leiden Communities

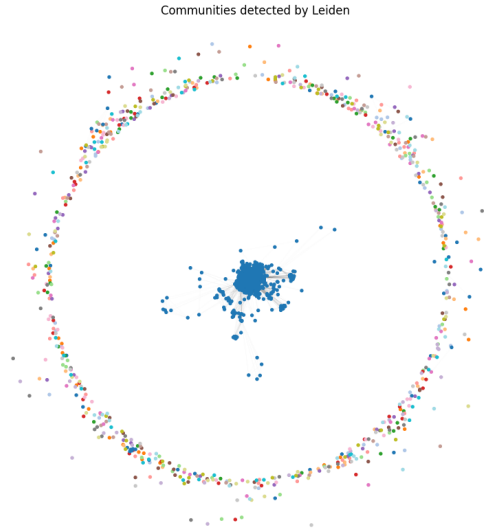


Figure 5: Communities detected by the Leiden algorithm.

7.3 Girvan–Newman Communities

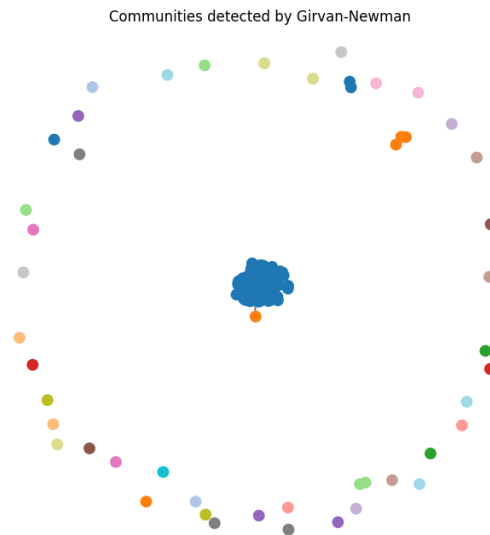


Figure 6: Communities detected by Girvan–Newman on the reduced graph.

8 Conclusions

This work investigated the community structure of the anime similarity network using Louvain, Leiden, and Girvan–Newman community detection algorithms.

The results show that Louvain and Leiden identified a highly consistent community structure, as indicated by the strong agreement measured by NMI and ARI. Their community size distributions were highly uneven, with a few large communities and many small ones. Girvan–Newman provided an additional perspective on community detection, but its application to a reduced subgraph limits the direct comparison with the other two algorithms.

Overall, Louvain and Leiden produced comparable community structures, while Leiden offered greater computational efficiency on the complete network.